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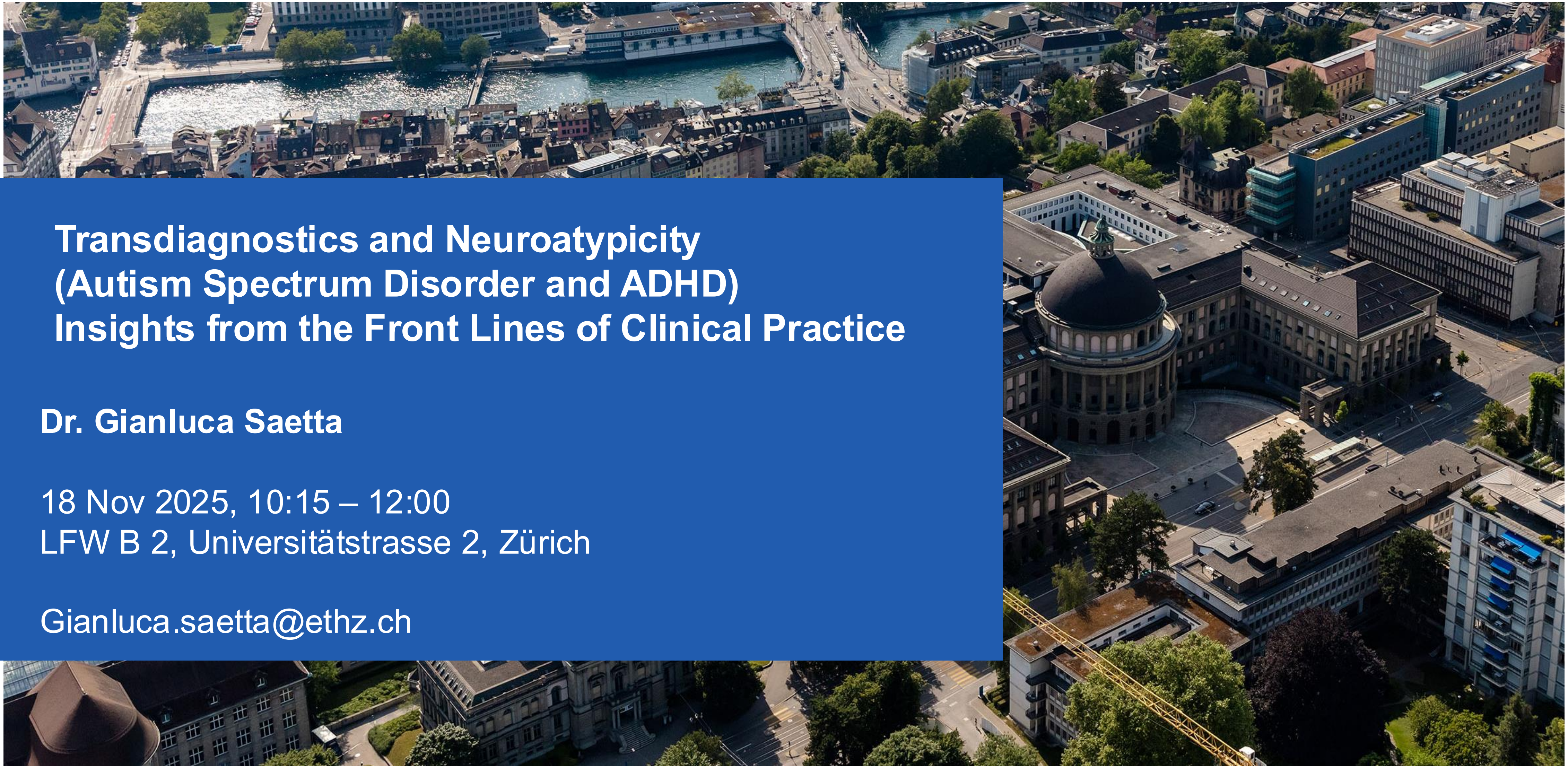


Transdiagnostics and Neuroatypicity (Autism Spectrum Disorder and ADHD) Insights from the Front Lines of Clinical Practice

Dr. Gianluca Saetta

18 Nov 2025, 10:15 – 12:00
LFW B 2, Universitätstrasse 2, Zürich

Gianluca.saetta@ethz.ch

ETH zürich

Overview of the lecture

1. **Autismus spectrum Disorder**

2. **The transdiagnostic approach**

3. **Translating it to practice: working with the ~~patient~~ person with a condition**

Example from clinical practice

Early markers of autism spectrum disorder

- Lack of response to name by 9–12 months of age
- Limited or absent eye contact and reduced social smiling
- Failure to use gestures (e.g., waving, pointing) by 12–18 months
- Poor imitation of others and lack of pretend play
- Reduced sharing of enjoyment or interests with caregivers
- Delayed or atypical language development, including echolalia or loss of previously acquired words
- Repetitive motor movements (e.g., hand flapping, finger flicking, body rocking) and clumsiness/rigidness
- Lining up toys or objects, and distress when the order is changed
- Atypical reactions to sensory stimuli (e.g., indifference to pain, adverse response to certain sounds or textures)
- Gender Identity fluctuations

American Family Physician | Issues | AFP By Topic | Collections | CME Quiz | Blog

← PREV ARTICLE | NOV 15, 2020 | NEXT ARTICLE →

PRACTICE GUIDELINES

Autism Spectrum Disorder: Updated Guidelines from the American Academy of Pediatrics

Home | JAMA | Vol. 329, No. 2

Review

Autism Spectrum Disorder
A Review

Tomoya Hirota, MD[1]; Bryan H. King, MD, MBA[1]

» Author Affiliations | Article Information

RELATED ARTICLES

Age-related changes in autism spectrum disorder

Reciprocal social interaction → often improves in adolescence/adulthood

Nonverbal communication → partial improvement, subtle cues often remain impaired

Restricted, repetitive behaviours → fairly stable; motor rituals ↓, sensory issues persist

Language/communication → often improve, especially after early delay

Friendships → difficulties in forming and keeping friends usually persist

Waizbard-Bartov E, Ferrer E, Heath B, et al. 2022

Autism: Diagnostic Criteria

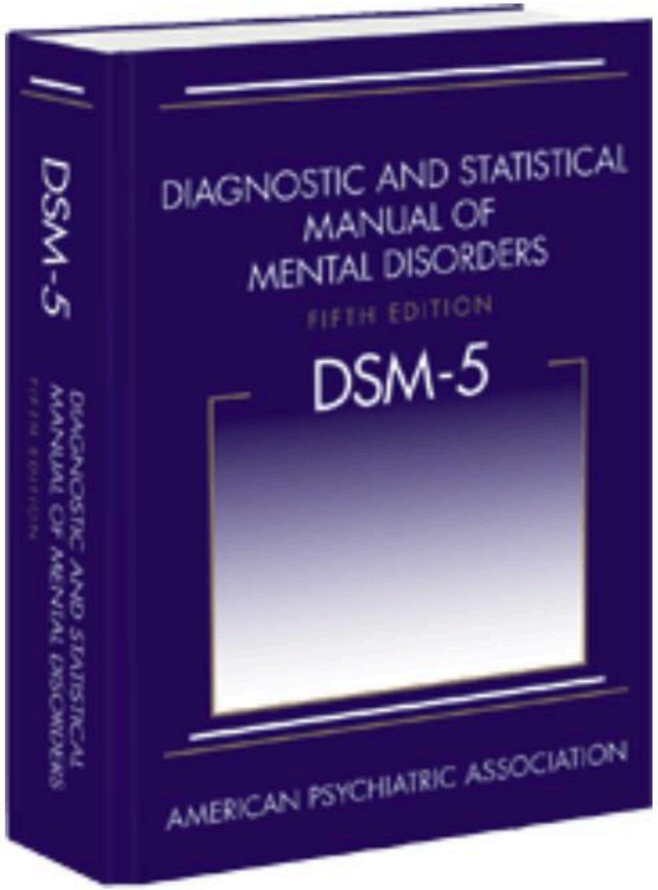
A. Persistent deficits in social communication and social interaction across multiple contexts. This includes deficits in social-emotional reciprocity, nonverbal communicative behaviors, and developing, maintaining, and understanding relationships.

B. Restricted, repetitive patterns of behavior, interests, or activities. Manifestations may include stereotyped or repetitive motor movements, insistence on sameness, highly restricted interests, or hyper-/hyporeactivity to sensory input.

C. Symptoms must be present in the early developmental period, but may not become fully manifest until social demands exceed limited capacities, or may be masked by learned strategies.

D. Symptoms cause clinically significant impairment in social, occupational, or other important areas of functioning

E. Disturbances are not better explained by intellectual disability or global developmental delay, social communication deficits must exceed those expected for general developmental level.



Towards the so-called Autismus Spectrum Disorder

Autism spectrum disorder (ASD) is currently defined as a single diagnostic category encompassing a range of presentations, including what was previously termed Asperger syndrome

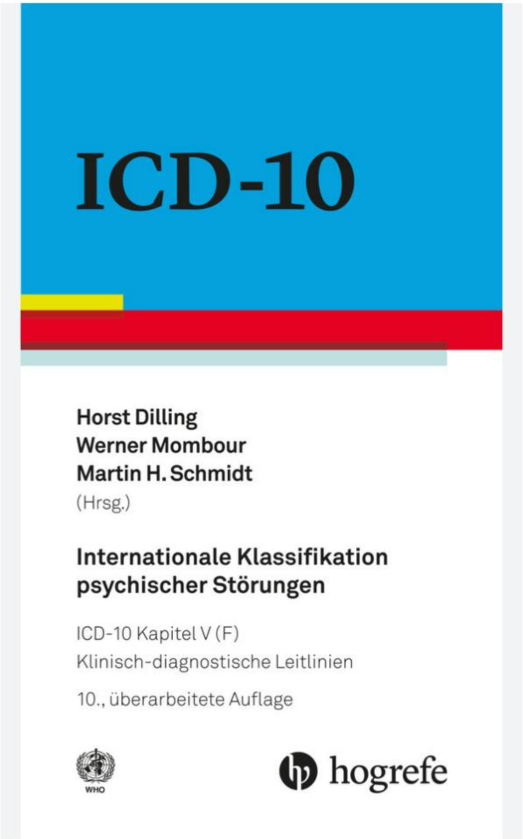
- **Key differences (historical):**

- Asperger syndrome: No significant language delay, no intellectual disability, higher verbal IQ, milder social impairment

- Autism: May include language delay, intellectual disability, broader range of severity.

The distinction is now **primarily historical**; clinical management and diagnosis are based on individual symptom profile and level of support needed, rather than subtype

ICD X stills differenciate between *Childhood autism, Atypical autism and Asperger Syndrome*



Autismus Spectrum Disorder

Home | JAMA Network Open | Vol. 7, No. 10

Original Investigation | Psychiatry

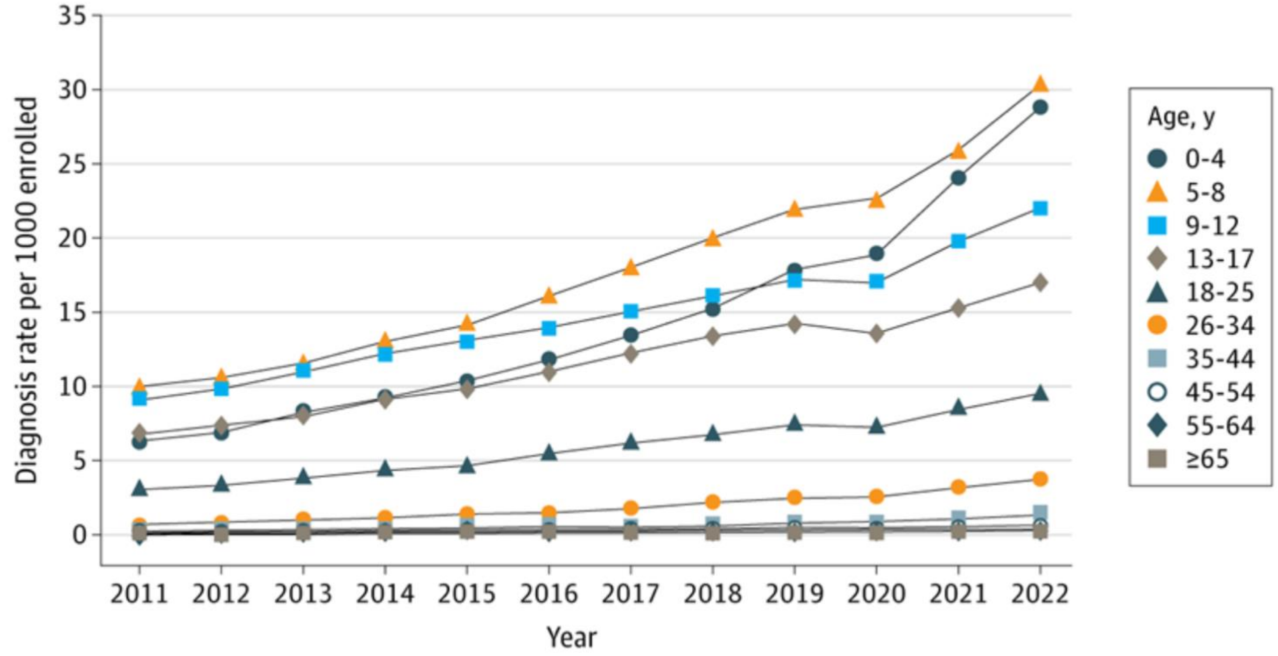
Autism Diagnosis Among US Children and Adults, 2011-2022

Luke P. Grosvenor, PhD[1]; Lisa A. Croen, PhD[1,2]; Frances L. Lynch, PhD[3] ; et al

» Author Affiliations | Article Information

RELATED ARTICLES FIGURES SUPPLEMENTAL CONTENT

Figure 3. Annual Autism Diagnosis Rates Among Members at MHRN Sites From 2011 to 2022, Stratified by Age Group



A total of 12 264 003 members were enrolled in 2022 (2 359 359 children aged 0 to 17 years [19.2%]

Autism affects about 3–4% of children and adolescents in the US, and about 1% of the global population

Prevalence of autism in Europe and Switzerland is approximately 0.7–1.5%, (0.48% in South-East France to 3.13% in Iceland)

Autism is diagnosed more frequently in males than females, with a male-to-female ratio of approximately 3–4:1.

Autismus Spectrum Disorder: Gene > Environment

Autism spectrum disorder (ASD) has a strong genetic component

Approximately 70% to 90% in large population-based studies and meta-analyses of twin and family cohorts.

Current clinical consensus recommends genetic testing (chromosomal microarray and fragile X testing, FMR1 gene) for all children diagnosed with ASD

JAMA Psychiatry

Home | Issues | Multimedia

Home | JAMA Psychiatry | Vol. 76, No. 10

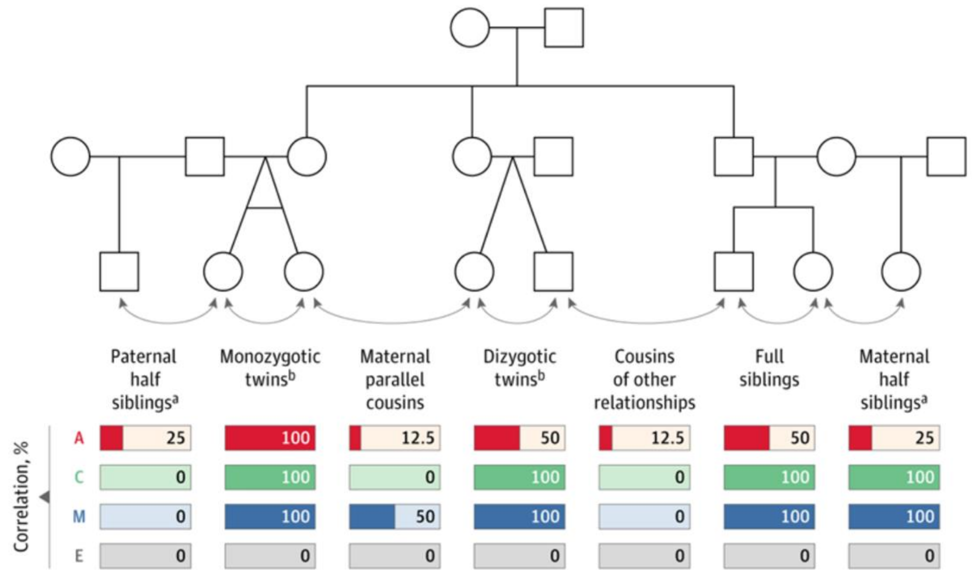
Original Investigation FREE

Association of Genetic and Environmental Factors With Autism in a 5-Country Cohort

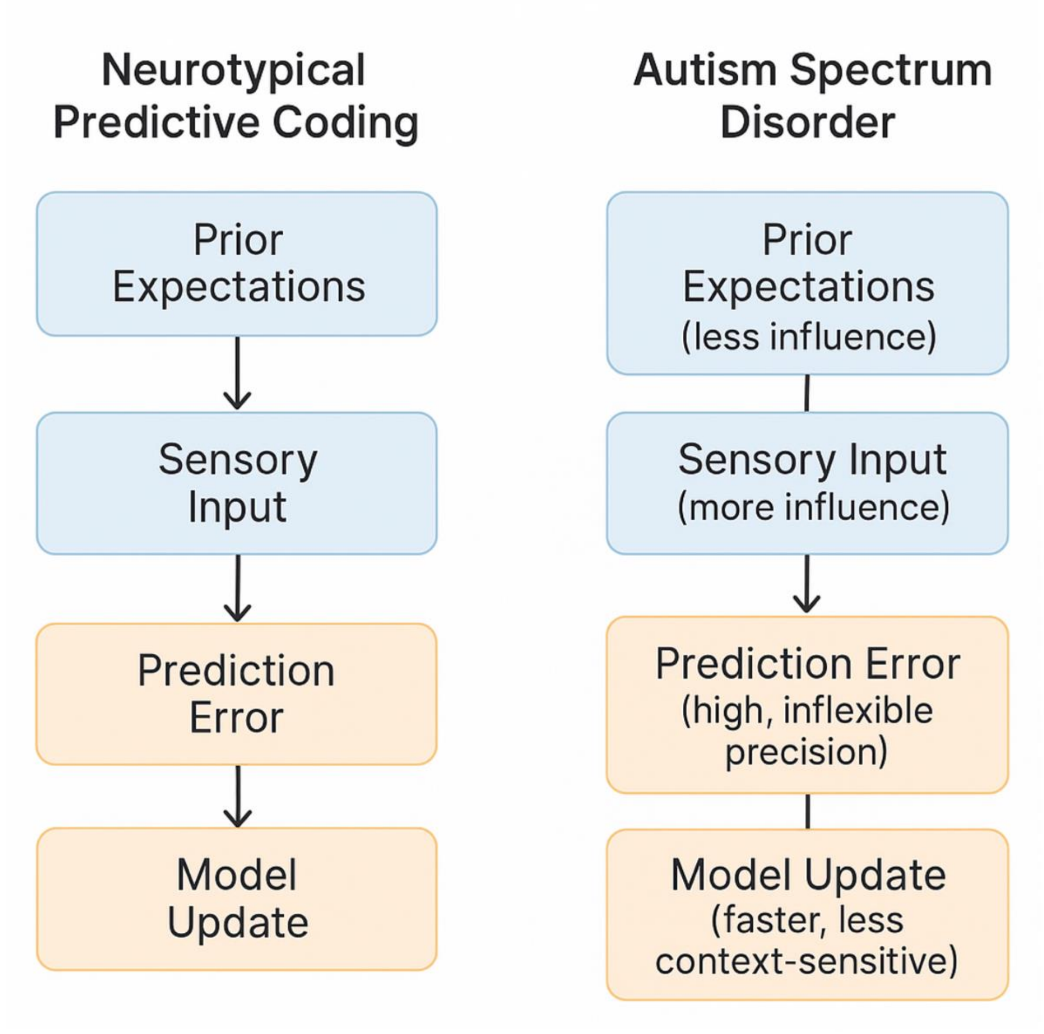
Dan Bai, MSc[1]; Benjamin Hon Kei Yip, PhD[1,2]; Gayle C. Windham, PhD, MSPH[3] ; et al

Author Affiliations | Article Information

Figure 1. Variance Component Derivation From Correlations Between Family Members



Autismus Spectrum Disorder: Predictive coding



Neurotypical in a café
• Has a rough picture in mind: "Cafés are noisy, that's fine."
• Ongoing noise (voices, cups, coffee machine) fits this picture, so most sounds are filtered as "background."
• Only something clearly unexpected, like a plate smashing, sticks out as a real "error" and briefly updates the situation: "Ah, someone dropped a plate."

Autistic in a café
• Relies less on the general idea "cafés are noisy" and more on each sound as it happens.
• Many sounds voices, cutlery, coffee machine, door, humming lights feel salient and less like "background."
• More sounds are treated as "errors" or small surprises, leading to constant updating of the situation and a higher risk of overload.

| Higher Neural Functions and Behavior

Predictive coding in autism spectrum disorder and attention deficit hyperactivity disorder

Authors: Maria Luz Gonzalez-Gadea, Srivas Chennu, Tristan A. Bekinschtein, Alexia Rattazzi, Ana Beraudi, Paula Tripicchio, Beatriz Moyano, ... SHOW ALL ... , and Agustin Ibanez | AUTHORS INFO & AFFILIATIONS

Publication: Journal of Neurophysiology • Volume 114, Issue 5 • https://doi.org/10.1152/jn.00543.2015

This is the final version - click for previous version

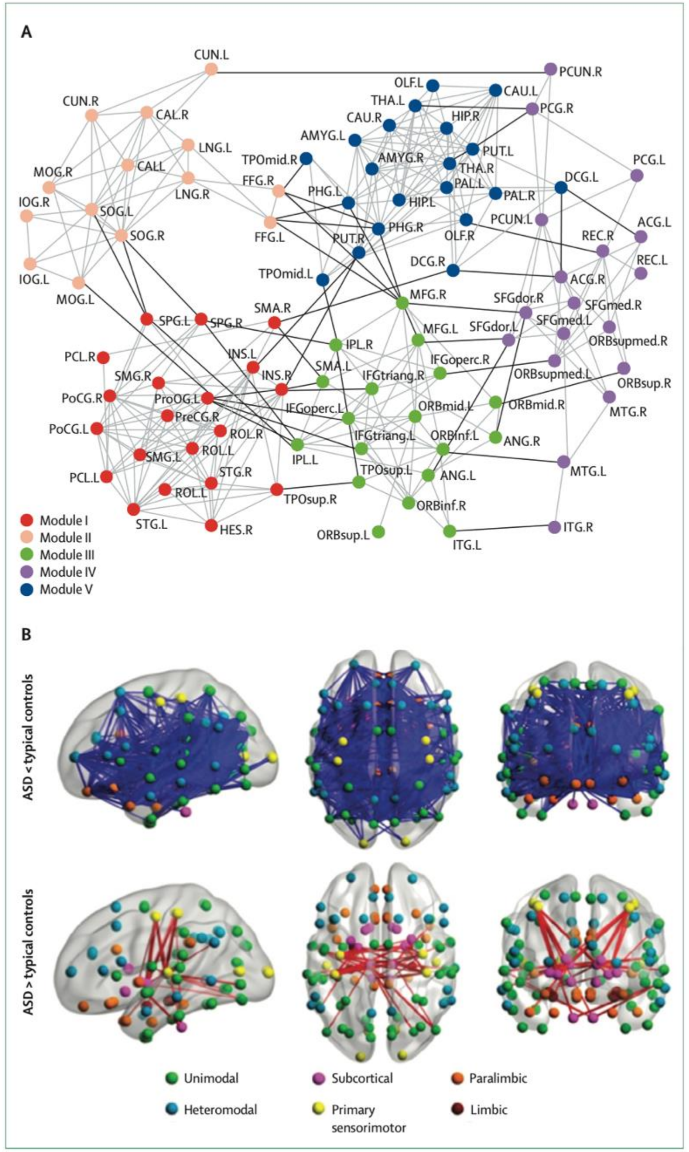
Autismus Spectrum Disorder: Neural markers

Early brain overgrowth
- ↑ Total brain volume between 6–24 months, especially frontal and temporal lobes
- By school age, growth slows and volumes approach those of typically developing peers

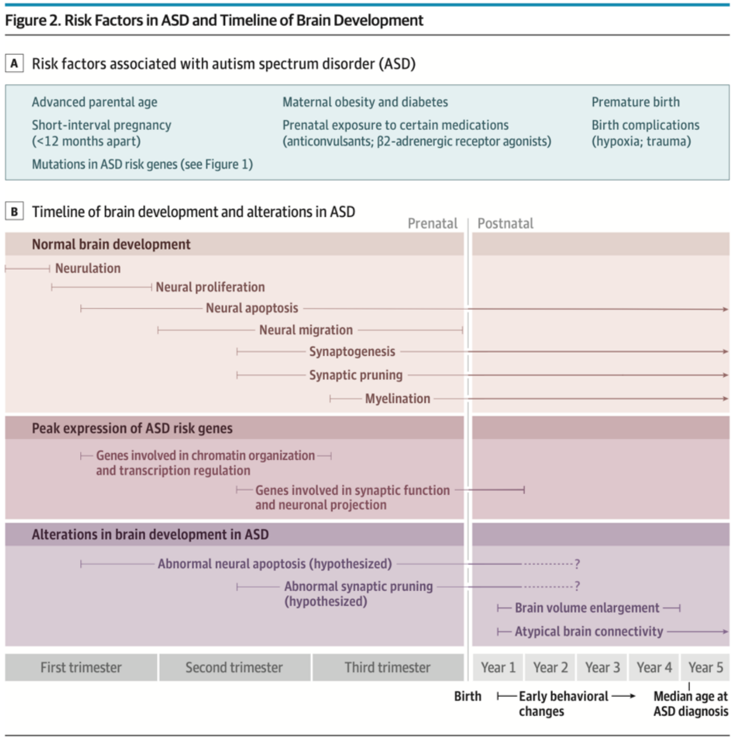
Atypical connectivity
- Reduced long-range connectivity (especially networks for social cognition and executive function)
- Increased local connectivity in regions such as frontal and occipital cortex

Regional differences
- Alterations in PCC, precuneus, dl/vlPFC, STS, fusiform face area, insula, amygdala, cingulate cortex
- These regions support social communication, sensory processing, and emotion regulation

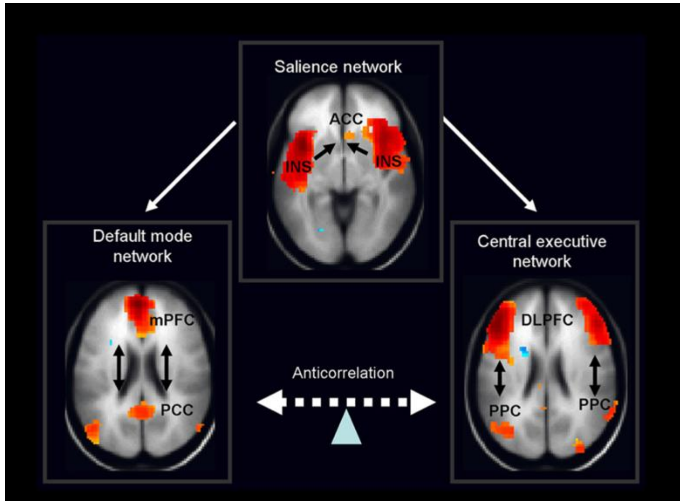
Triple Network
- Less connectivity within DMN/SN, more connectivity between DMN–SN and DMN–ECN
- this disrupts flexible switching between internal and external attention, social communication problems, executive dysfunction, and restricted, repetitive behaviors



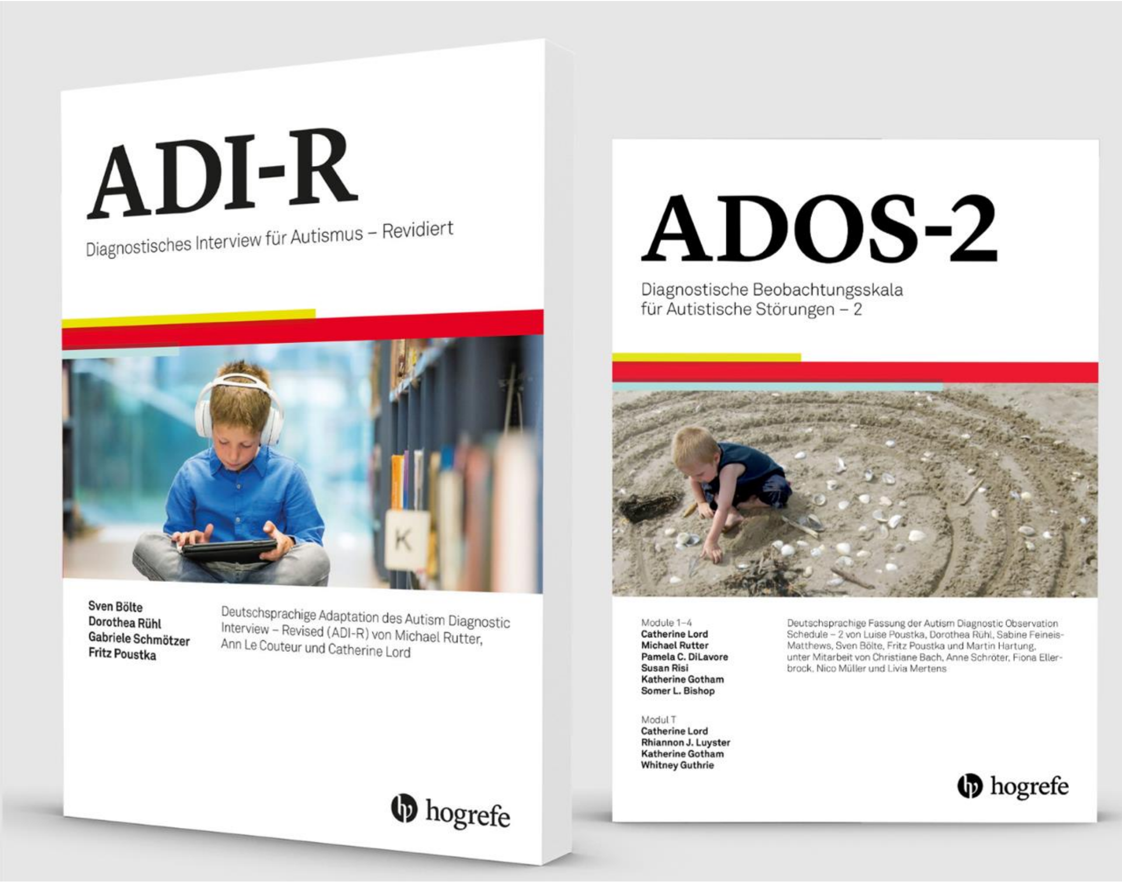
Ecker et al., Lancet Neurol 2015



Muhle et al., Jama Psy. 2018



Autismus Spectrum Disorder: Diagnostic



◆ Patriquin et al. ◆

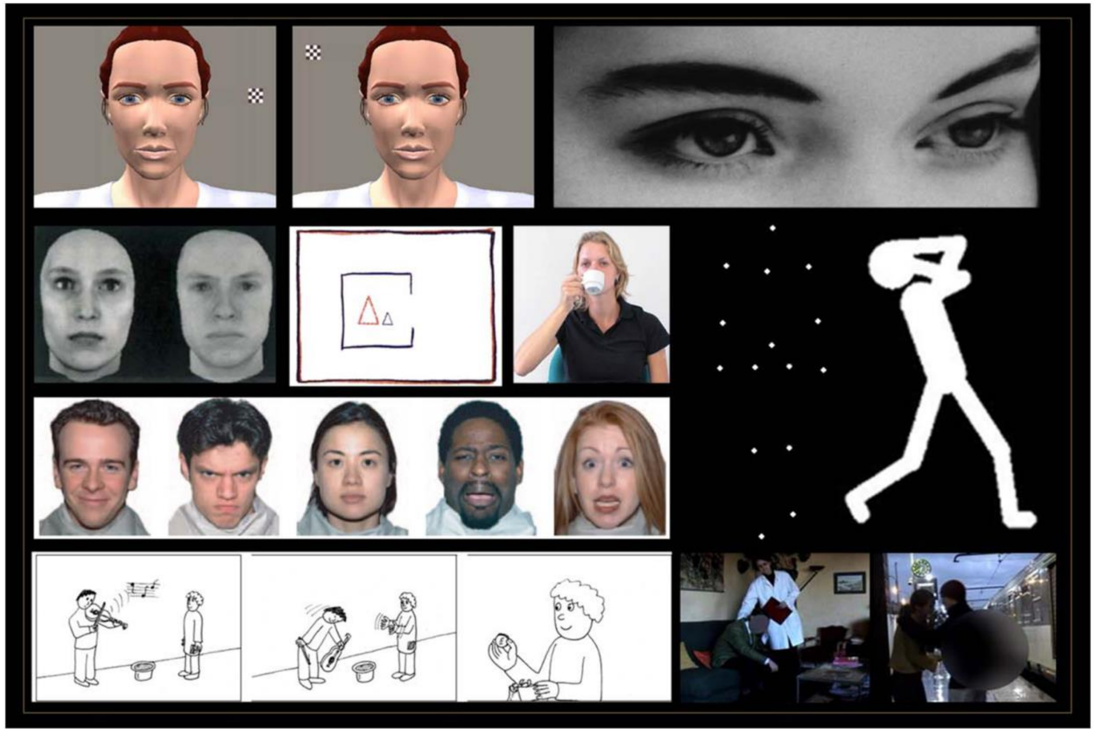


Figure 1.
Examples of social cognition tasks used in studies included in the meta-analysis. [Color figure can be viewed at wileyonlinelibrary.com]

Autismus Spectrum Disorder: Treatment

There is no cure for ASD!!!; treatment is individualized based on symptoms, comorbidities, and functional needs

Medications
•Does not treat core ASD symptoms
•Risperidone and aripiprazole for irritability and aggression
•Stimulants for comorbid ADHD
•Experimental biomedical treatments **are not recommended!**

Evidence for use of medication in autism spectrum disorder

	Age (years) for use as indicated by US FDA	Target symptoms	Effect size (d)	Common adverse effects
Risperidone	5–16	Agitation or irritability in ASD	0.94[123]	Increased appetite, sedation, weight gain
Aripiprazole	6–17	Agitation or irritability in ASD	0.87[124]	Nausea, weight gain
Atomoxetine	6–15	Typically for ADHD symptoms	0.68–0.84[129]	Decreased appetite nausea, irritability
Methylphenidate	≥6	ADHD	−0.78 (95% CI −1.13 to −0.43) (teacher-rated)[128]	Sleep disruption, decreased appetite
Guanfacine	6–12	ADHD	1.67[130]	Fatigue, sedation, decrease in pulse and blood pressure

ASD=autism spectrum disorder. ADHD=attention-deficit hyperactivity disorder. FDA=US Food & Drug Administration.

Lord C, Elsabbagh et al., *Lancet*. 2018

Autismus Spectrum Disorder: Treatment

There is no cure for ASD!!!; treatment is individualized based on symptoms, comorbidities, and functional needs

Realistic Cogntive Behavioral Therapy Goals

- Reduce anxiety, depressive symptoms, OCD like problems and anger
- Improve emotion regulation, stress coping and tolerance of uncertainty
- Decrease avoidance and increase participation in daily life (school, work, social situations)
- Build concrete skills for problem solving, planning and behavior regulation
- Improve recognition of own emotions, thoughts and early warning signs of overload

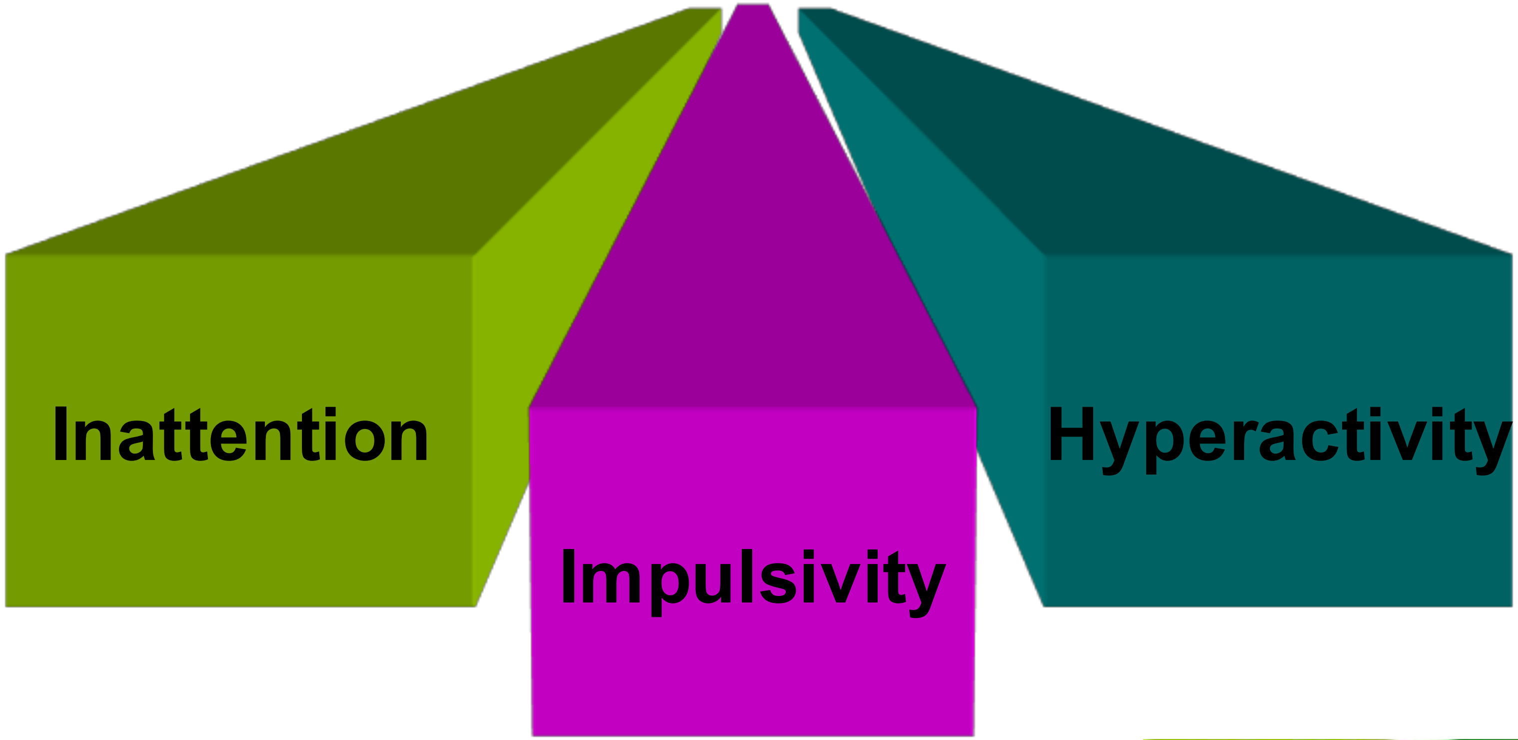
Guidelines for successfull therapy:

- highly structured, concrete, visually supported CBT with clear, literal language
- emotion recognition and regulation training, often including work on alexithymia and using personal interests as examples
- Adapt setting and pacing: predictable routines, reduced sensory load, breaks, repetition, and slower skill acquisition if needed
- Concrete language
- Involvement of caregiver or support persons

Saetta et al., in preparation; Lord C, Elsabbagh et al., *Lancet*. 2018

WHAT IS ADHD?

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental ~~disorder~~ **CONDITION** characterized by a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning or development.



WHAT IS ADHD?

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Inattention	Impulsivity	Hyperactivity
• Has difficulty paying attention • Tends to avoid tasks that require sustained effort • Often appears not to listen when spoken to directly • Frequently leaves tasks unfinished • Struggles to stay organised • Often misplaces things and seems forgetful • Is easily distracted by surrounding events or stimul	• Fidgets or squirms • Gets up from seat when expected to remain seated • Runs or climbs in situations where it is not appropriate • Finds it hard to play or work quietly • Seems to be constantly on the move • Talks a lot	• Blurts out answers before questions are finished • Has difficulty waiting for their turn • Frequently interrupts others • Intrudes on or interferes with others' activities

ADHD: Diagnostic Criteria

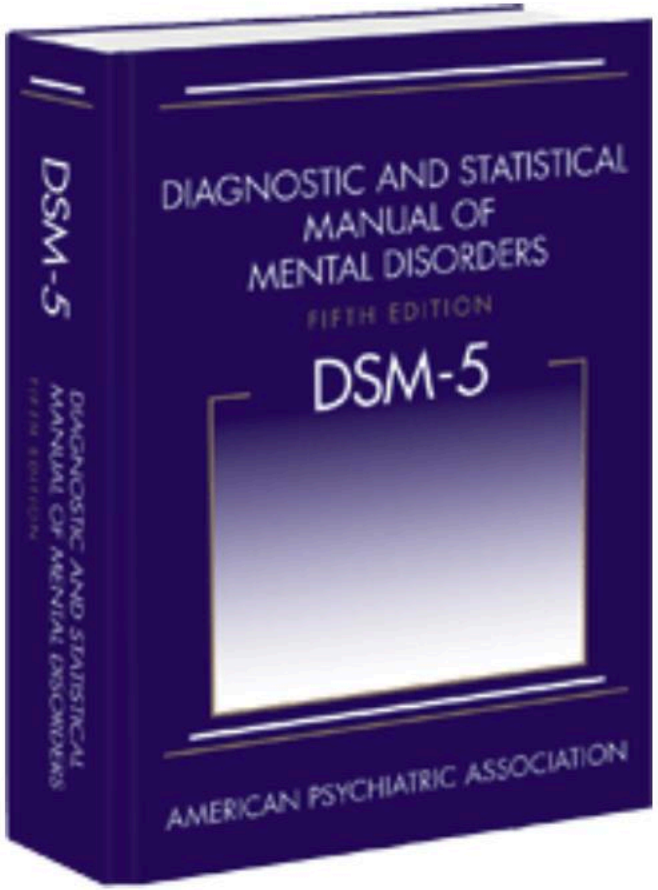
A Persistent pattern of inattention and/or hyperactivity-impulsivity, as characterized by at least **six symptoms** (or five for individuals aged 17 and older) of inattention and/or hyperactivity-impulsivity, persisting for at **least six months** to a degree inconsistent with developmental level and negatively impacting social, academic, or occupational activities.

B. Several inattentive or hyperactive-impulsive symptoms must have been present **prior to age 12 years.**

C. Symptoms must be present in two or more settings.

D. There must be clear evidence that symptoms interfere with, or reduce the quality of, social, academic, or occupational functioning.

E. The symptoms are not better explained by another mental disorder or exclusively during the course of another disorder.



ADHD: Epidemiology in Europe and Switzerland

THE LANCET

This journal | Journals | Publish | Clinical | Global health | Multimedia | Events

SEMINAR · Volume 395, Issue 10222, P450-462, February 08, 2020 · Download Full Issue

Attention-deficit hyperactivity disorder

Jonathan Posner, MD [a,b] · Guilherme V Polanczyk, MD [c] · Edmund Sonuga-Barke, PhD [d]

Affiliations & Notes · Article Info

Prevalence
- Children/adolescents: ~2.9% (Europe, community studies)
- Adults: ~2.5%
- Male predominance: male:female ≈ 2–4:1

Trends in Europe
- Strong increase in diagnoses and ADHD medication, especially in adults and female
- Denmark: 0.10% (2000) → 3.03% (2022), largest rise in young adult women
- Sweden: ADHD medication use ↑ ~5× in children and >10× in adults (2006–2020)

Drivers of rising diagnoses/medication
- Broader and more **«flexible»** use of DSM/ICD criteria, especially for adults and females
- Greater public, school and professional awareness **(?)**

Additional points

Marked geographical variability: higher rates in Northern vs Southern/Western Europe

COVID-19: some fluctuation in incidence, but no sustained increase in overall prevalence since 2020

No relationship to vaccines!

ADHD: Gene > Environment

- Heritability of ADHD ≈ 70–80%, including in European and Swiss cohorts.
- Genetic contribution has stayed stable over time; rising diagnosis/medication rates reflect awareness and practice, not changing genes.
- Environmental factors (for example prenatal stress, prematurity, severe deprivation) add risk on top of genetics.
- Gene–environment interactions are plausible but evidence is still limited and inconsistent
- Candidate gene studies, particularly those involving dopamine-related genes (e.g., DRD4, DAT1, MAOA),

Neuroscience and Biobehavioral Reviews 139 (2022) 104757

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journal homepage: www.elsevier.com/locate/neubiorev

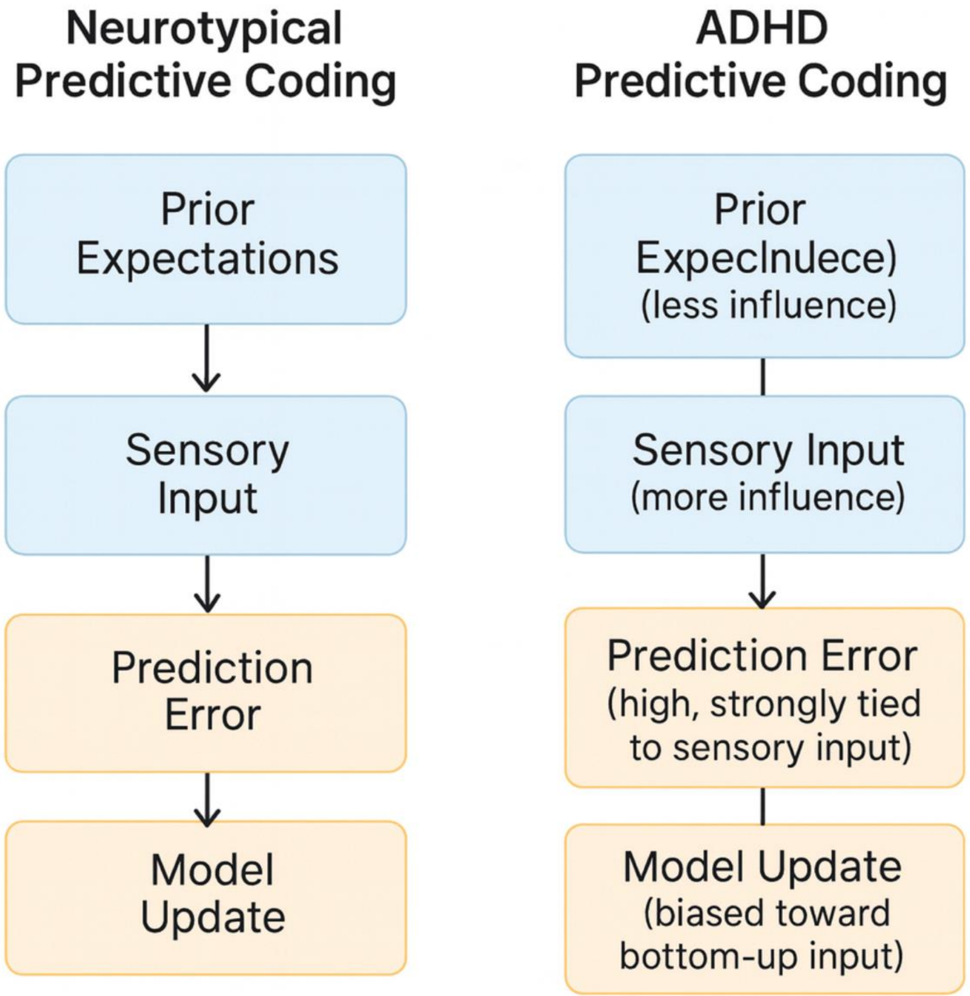
ELSEVIER

ADHD co-morbidities: A review of implication of gene × environment effects with dopamine-related genes

Margus Kanarik[a], Oliver Grimm[b], Nina Roth Mota[c,d], Andreas Reif[b], Jaanus Harro[a,e,*]

[a] Chair of Neuropsychopharmacology, Institute of Chemistry, University of Tartu, Ravila 14A Chemicum, 50411 Tartu, Estonia
[b] Department of Psychiatry, Psychosomatic Medicine and Psychotherapy, University Hospital, Goethe University, Frankfurt, Germany
[c] Department of Human Genetics, Donders Institute for Brain, Cognition and Behaviour, Radboud University Medical Center, Nijmegen, the Netherlands
[d] Department of Psychiatry, Donders Institute for Brain, Cognition and Behaviour, Radboud University Medical Center, Nijmegen, the Netherlands
[e] Psychiatry Clinic, North Estonia Medical Centre, Paldiski Road 52, 10614 Tallinn, Estonia

ADHS: Predictive coding



Neurotypical in a café
• Has a rough picture in mind: “Cafés are noisy, that’s fine.”
• Ongoing noise (voices, cups, coffee machine) fits this picture, so most sounds are filtered as “background.”
• Only something clearly unexpected, like a plate smashing, sticks out as a real “error” and briefly updates the situation: “Ah, someone dropped a plate.”

ADHD in a café
• Comes in with the idea “cafés are noisy,” but what is happening right now quickly takes over.
• Attention keeps jumping to each new or slightly different sound: a spoon hits a cup, the door beeps, the barista calls a name, a phone vibrates.
• Each small change feels like a little “something is happening” signal, so focus is pulled away again and again, the brain keeps rewriting the scene, and it becomes hard to stay with one conversation or train of thought..

Autism --> “too many things are relevant at the same time,” leading to overload.
ADHD --->“what is relevant keeps jumping around,” leading to scattered attention.

ADHS: Neural markers

Dopamine dysregulation

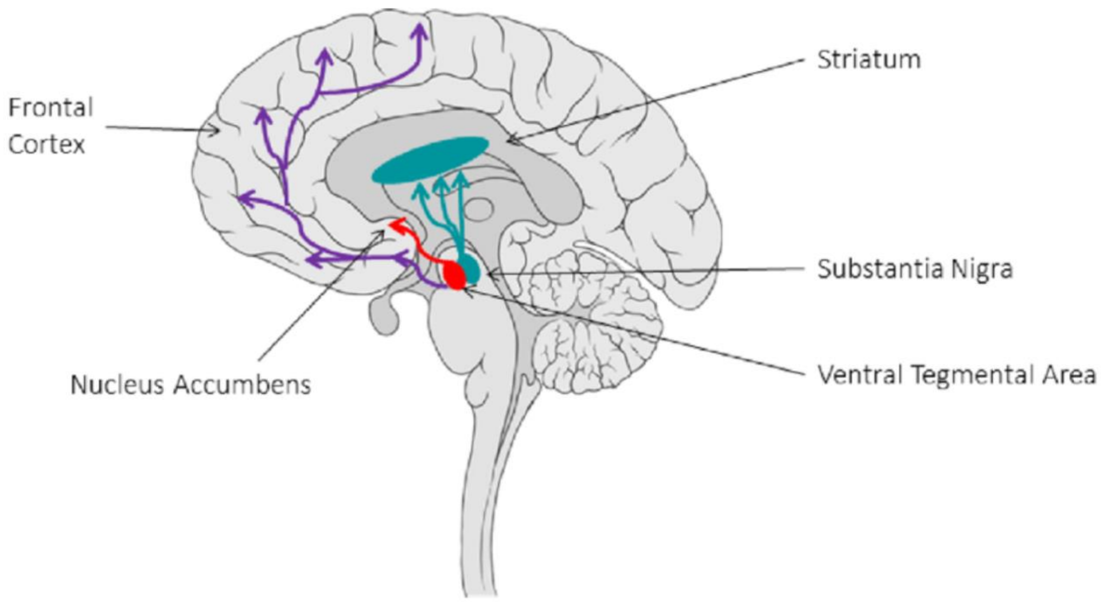
- Reduced tonic and increased phasic dopamine in striatum (especially caudate)
- Altered DAT and D2/D3 receptor availability in caudate, nucleus accumbens and midbrain
- Unstable reward signalling → inattention, impulsivity, higher addiction risk

Neurodevelopment and circuits

- Delayed cortical maturation in prefrontal cortex
- Reduced volume and altered function in frontostriatal circuits and basal ganglia
- Leads to weak inhibitory control, planning and motivation

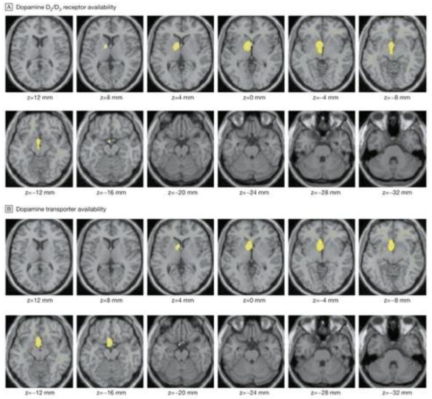
Triple Network

- DMN: insufficient task related deactivation → mind wandering, lapses of attention
- ECN/FPN: hypoactivation and weaker connectivity → poor top down control
- SN: inefficient switching between DMN and ECN → distractibility and impulsive decisions

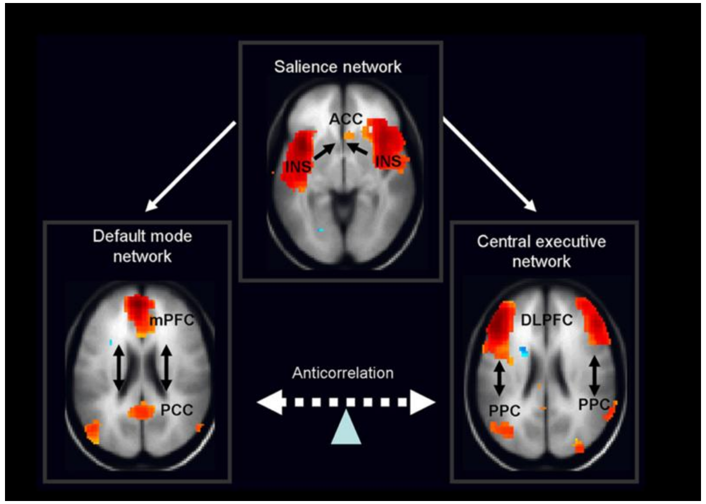


Schematic representation of the major dopaminergic pathways in the brain: the nigrostriatal pathway (blue), the mesolimbic pathway (red) and the mesocortical pathway (purple).

Reneman et al., 2021. European Journal of Radiology



Volkow et al., 2009, Jama



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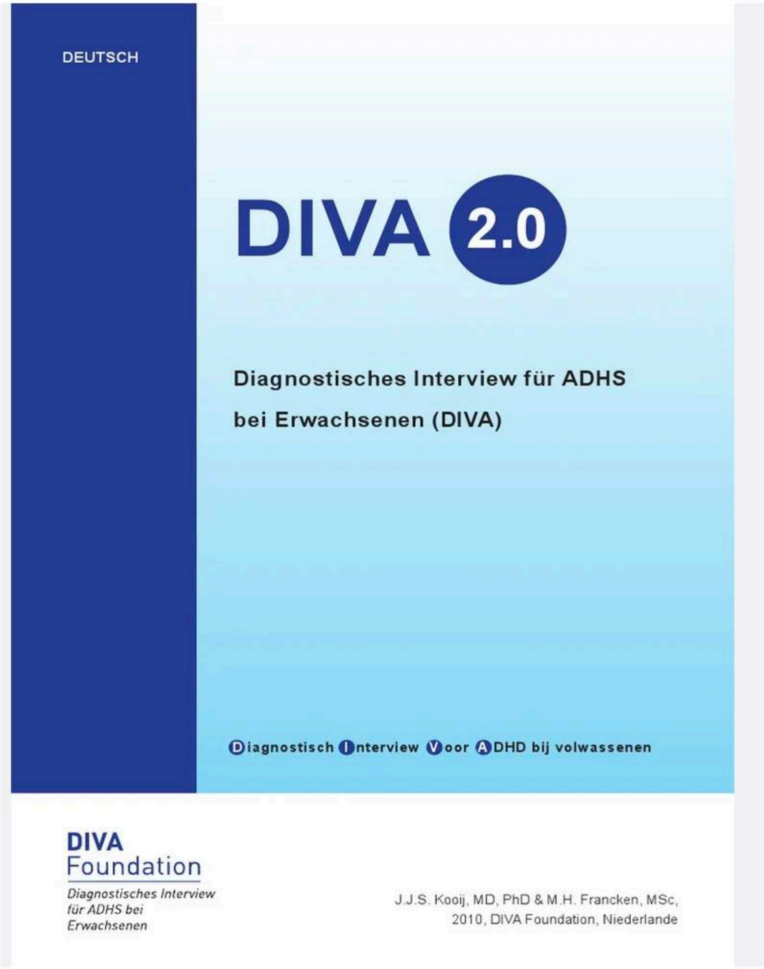
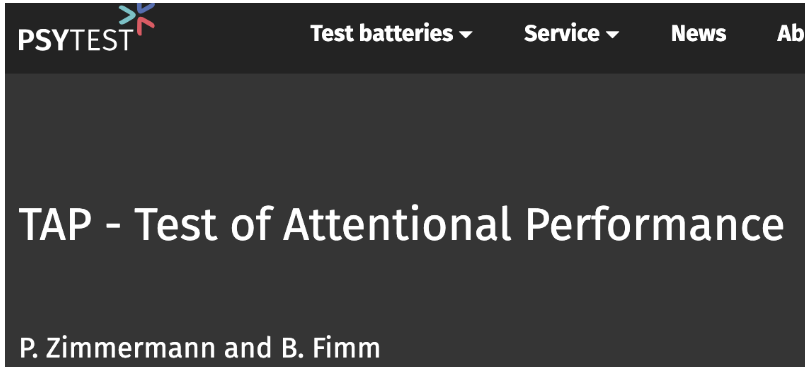
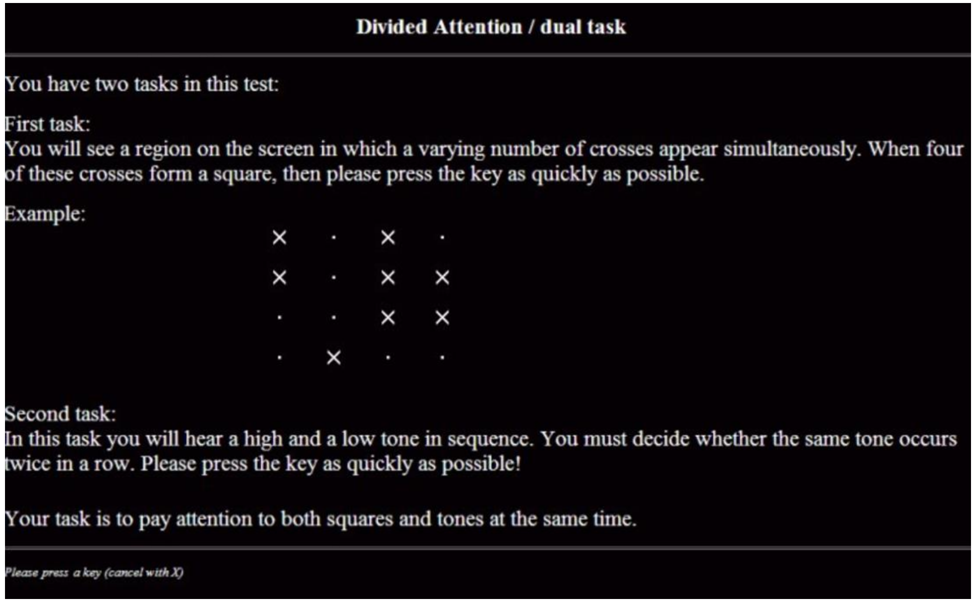
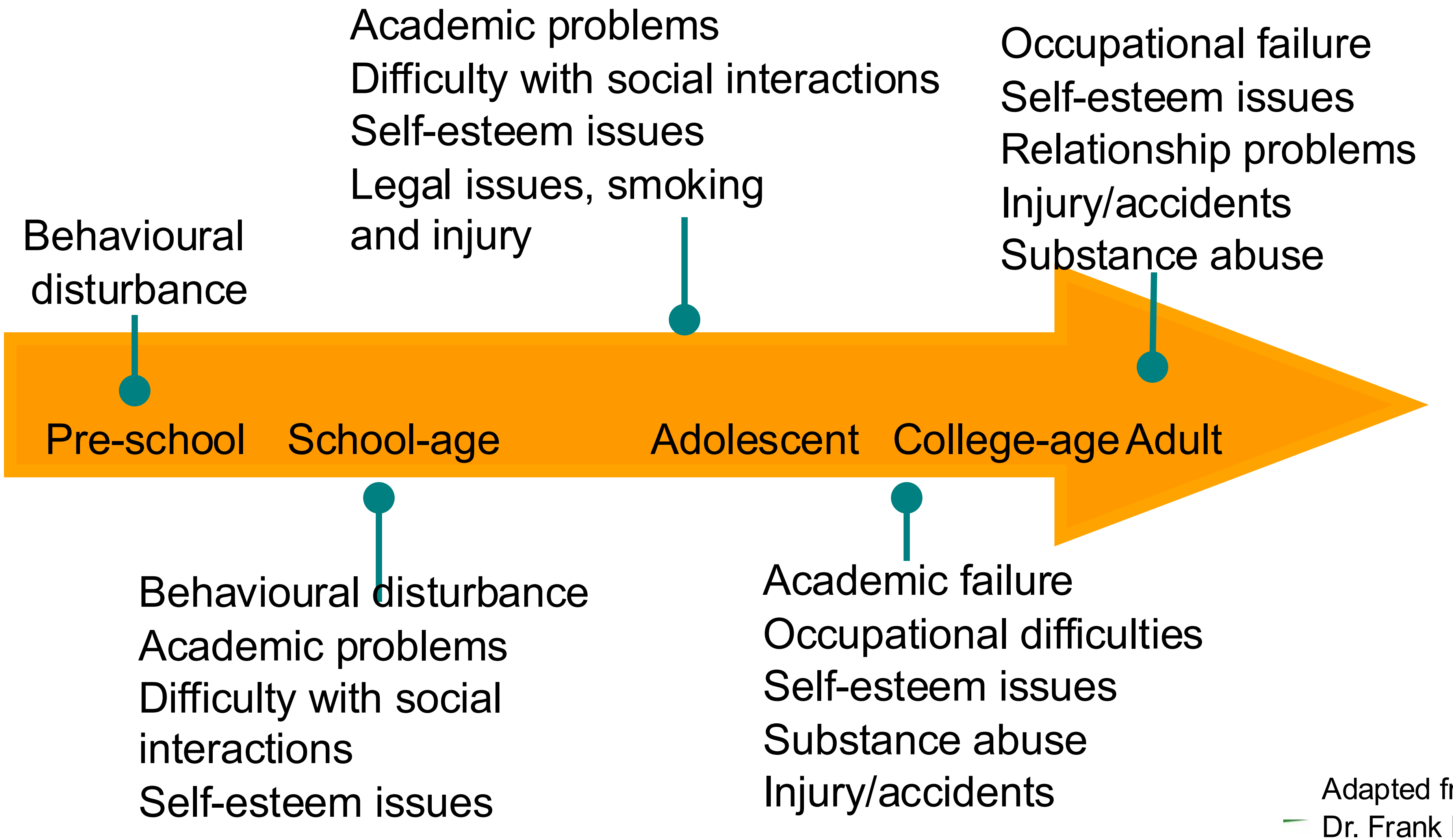
ADHS: Diagnostic







ADHS: Developmental Impact of ADHD



Adapted from presentation from
Dr. Frank Röhricht

East London
NHS Foundation Trust

Autismus Spectrum Disorder: Treatment

Realistic Cognitive Behavioral Therapy Goals

- Reduce anxiety, depressive symptoms, OCD-like problems and anger
- Improve emotion regulation, stress coping and tolerance of uncertainty
- Decrease avoidance and increase participation in daily life
- Build concrete skills for problem solving, planning and behavior regulation
- Improve recognition of own emotions, thoughts and early warning signs of overload

Guidelines for successful therapy

- Highly structured, concrete, visually supported CBT with clear, literal language
- Focus on core tools --> psychoeducation, cognitive restructuring, exposure/behavioral activation, problem-solving, affect regulation
- Use visual schedules, checklists and emotion charts to support understanding and practice
- Adapt setting and pacing --> predictable routines, reduced sensory load, breaks, repetition, slower pacing if needed
- Emotion recognition and regulation training, including work on alexithymia and use of personal interests as examples
- Involvement of caregivers or support persons in homework, everyday practice and monitoring

Saetta et al., in preparation; Lord C, Elsabbagh et al., *Lancet*. 2018

Neurodiversity and Transdiagnostic Dimensions
From Categorical Diagnosis to Person-Centred Neurodiversity

Neurodiversity
Autism, ADHD as natural cognitive variations
Emphasis on strengths, identity, wellbeing, and inclusion rather than “normalisation”

Limits of categorical diagnosis
Arbitrary boundaries, high heterogeneity and comorbidity
Misses subclinical difficulties and real support needs

Transdiagnostic dimensions
Continuous traits: attention, social communication, executive functions
Allow a more fine-grained individual profile

Data-driven profiles
Clustering reveals overlapping mechanisms and behavioural patterns
Diagnostic labels are scattered across subtypes, not tied to distinct biological entities

Person-centred practice
Target impairing traits, build on strengths
Provide accommodations in school, workplace, and healthcare settings

Miranda-Ojeda et al., (2025), *Developmental Neurobiology,* Chapman and Botha, (2023), *Developmental Medicine and Child Neurology,* Pellicano and den Houting, (2022), *Journal of Child Psychology and Psychiatry, and Allied Disciplines,* Sonuga-Barke, (2023), *Journal of Child Psychology and Psychiatry, and Allied Disciplines*

Translating it to practice: working with a person with a condition

Example from clinical practice

Interview and open discussion with Mr X

THANK YOU FOR YOUR ATTENTION!

Whether it was focused, wandering or hyperfocused on one slide

Your brain was absolutely invited ;)